Reeth S Kawad

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SKILLS

Test & Measurement: DMM, oscilloscope, LabVIEW, Python, C++, sensor calibration, hardware-in-the-loop (HIL), uncertainty
Electrical, Firmware & Controls: Python, Arduino, PID control, state machine design, sensor integration and debugging
Mechanical Design, **Fabrication & Integration**: SolidWorks, Siemens NX, GD&T, Haas CNC Mill/Lathe, soldering, rapid prototyping (FDM/SLA), hardware-software bring-up, vendor coordination

WORK EXPERIENCE

Robotics Systems and Applications Intern, GrayMatter Robotics *Jan 2026-Present*

- Built a **multimodal DAQ pipeline** integrating force, IMU, position trackers, and thermal sensors for **hardware-in-the-loop (HIL)** characterization of human sanding processes, reducing time study effort by 40%
- Redesigned a universal sanding fixture across 14 SKUs after diagnosing vision-segmentation failure; introduced spring-loaded compliant features and displacement sensors to validate part seating before each test cycle
- Ran iterative force testing to identify failure modes and quantify performance across iterations for production quality

USC Baum Family Makerspace, *Machinist and Fabrication Engineer* *Sep 2023-Dec 2025*

- Machined **GD&T** and precision (±1 thou) components on Haas CNC Mill/Lathe, ProtoTRAK, and Omax waterjet; verified against **technical drawings** and optimized toolpaths in **MasterCam**
- Supported 12+ design teams and 80+ research labs with design-for-manufacturing (**DFM**) guidance and operate 3D printers, reduce iteration cycles by catching manufacturability issues at the design stage

Mechanical Engineer, USC Dynamic Robotics and Controls Laboratory *Sep 2025-Dec 2025*

- Designed and rapid-prototyped a **series elastic actuator (SEA**) for an N20 motor as part of an 8-DOF robotic hand, including spring selection and mounting, and parallel link joint assembly in **SolidWorks** for rigid-actuation finger
- Performed torque calculations for **motor** selection and validated output torque using bench-level force sensor measurements
- Simulated hand kinematics/dynamics in **MuJuCo** to validate design decisions before hardware build

Mechanical Engineering Intern, Lumindt Labs *Jun 2025-Aug 2025*

- Designed and fabricated a thermal test fixture in **SolidWorks** housing a transient hot-wire sensor; wrote **Python DAQ (Raspberry Pi)** to measure thermal conductivity of four metal powders (<10% uncertainty), enabling material down-selection
- Performed BOP component selection and ASME BPVC-based stress analyses to validate structural integrity under pressure

LEADERSHIP

Founder & Team Lead, Electronics and Controls, Collegiate Wind Competition *Sep 2025-May 2026*

- Designed an **electric braking system** using MOSFETs and power resistors as a variable resistive load for rated-power control **and** safe turbine shutdown
- Built full integrating RPM. Voltage and current sensors via I2C on an Arduino UNO; validated control loop against expected electrical characteristics using an oscilloscope and multimeter
- Performed hands-on soldering, hardware bring-up and sensor debugging to deliver a complete, working turbine controls architecture from scratch

Aerodynamics and Vehicle Dynamics Engineer, Formula SAE (SC Racing) *Jan 2023-Apr 2025*

- Developed a yaw probe and toe alignment tool to validate CFD simulations against physical measurements, and automated CFD post processing via MATLAB scripting

PROJECTS

Performance Analysis of Honeycomb Flow Straightener in a Wind Tunnel

- Designed in **Siemens NX** , fabricated using laser cutter, and instrumented a 1.5m blow-down wind tunnel to validate a honeycomb flow straightener, driving a **rack-and-pinion stepper motor** and pitot tube via **LabVIEW** and analyzing results in **MATLAB** - reducing turbulence intensity 53% at the inlet and 86% at the test section

EDUCATION

University of Southern California, Viterbi School of Engineering Los Angeles, CA
B.S. Mechanical Engineering *May 2026*
Relevant Coursework: Linear Control Systems, Turbine Design and Analysis, Heat Transfer, Mechanics of Materials, Fluid Dynamics